Day 010 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Model Multi-Step Decimal Decisions

Standards: 6.NS.B.3

Learning target: I can model Multi-Step Decimal Decisions, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Align decimal points and add placeholder zeros. Regroup one whole as ten tenths, or one tenth as ten hundredths. Subtract by place and check by adding the difference to the subtrahend. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 9

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01 / Learn

Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Align decimal points and add placeholder zeros. Regroup one whole as ten tenths, or one tenth as ten hundredths. Subtract by place and check by adding the difference to the subtrahend. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 18.3 + 6.8. Show your regrouping or division steps.

1. Line up the decimal points. Add placeholder zeros so the numbers are 18.3 and 6.8.

2. Column 1 from the right: 3 + 8 = 11. Write 1 and carry 1 to the next column.

3. Column 2 from the right: 8 + 6 + 1 carried = 15. Write 5 and carry 1 to the next column.

4. Column 3 from the right: 1 + 0 + 1 carried = 2. Write 2.

5. Bring the decimal point straight down, keeping 1 decimal place: 25.1. The simplified decimal answer is 25.1.

Answer: 25.1

Worked example 2

Use the standard algorithm: 3.32 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 83
B. 93/10
C. 83/10
D. 73/10

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 10: 33.2 ÷ 4 = 8.3.

3. Check: 0.4 × 8.3 = 3.32.

Answer: C - 83/10

Day 010 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 36.5 + 3.2. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 53.1 - 8.5. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 70.1 × 4.3. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 11.85 ÷ 1.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 6.99 + 4.9. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 3.65 - 3.3. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 2.55 × 0.51. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 1.68 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 26/5
B. 21/5
C. 42
D. 16/5

09 / Multiple choice

Use the standard algorithm: 7.27 + 0.85. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 178/25
B. 228/25
C. 203/25
D. 406/5

10 / Constructed response

A tank holds 6.96 L. A class uses 0.26 L. How much remains?

Day 010 | Unit 01 | Answer key

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1. 39.7

Align decimal points; 36.5 + 3.2 = 39.7.

6.NS.B.3 | Constructed response

2. 44.6

Align decimal points; 53.1 - 8.5 = 44.6.

6.NS.B.3 | Constructed response

3. 301.43

Compute the whole-number product and restore decimal places: 70.1 × 4.3 = 301.43.

6.NS.B.3 | Constructed response

4. 7.9

Multiply both numbers by 10: 118.5 ÷ 15 = 7.9. Check: 1.5 × 7.9 = 11.85.

6.NS.B.3 | Constructed response

5. 11.89

Align decimal points; 6.99 + 4.9 = 11.89.

6.NS.B.3 | Constructed response

6. 0.35

Align decimal points; 3.65 - 3.3 = 0.35.

6.NS.B.3 | Constructed response

7. 1.3005

Compute the whole-number product and restore decimal places: 2.55 × 0.51 = 1.3005.

6.NS.B.3 | Constructed response

8. B - 21/5

Multiply both numbers by 10: 16.8 ÷ 4 = 4.2. Check: 0.4 × 4.2 = 1.68.

6.NS.B.3 | Multiple choice

9. C - 203/25

Align decimal points; 7.27 + 0.85 = 8.12.

6.NS.B.3 | Multiple choice

10. 6.7 L

Align decimal points; 6.96 - 0.26 = 6.7.

6.NS.B.3 | Constructed response